FAISAL VANU

Software Developer | AI/ML Enthusiast
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PROFESSIONAL SUMMARY

Computer Science Engineering graduate specializing in Artificial Intelligence and Machine Learning from the University of Mumbai. Hands-on project experience across full-stack web development and applied AI/ML, working with React, FastAPI, PyTorch, and Flutter. Comfortable moving between building APIs and interfaces and training and evaluating models. Looking for an entry-level software development or AI/ML engineering role where I can keep building practical, working systems.

TECHNICAL SKILLS

Programming: Python, JavaScript, TypeScript, Dart, SQL
Frontend: React, Next.js, HTML, CSS, Tailwind CSS
Backend: FastAPI, Node.js, Express.js
Mobile: Flutter
AI/ML: PyTorch, Scikit-learn, LightGBM, YOLOv8, OpenCV, Pandas, NumPy, SHAP, Grad-CAM
Databases: PostgreSQL, SQLite, MongoDB, Supabase
Tools & Technologies: Git, GitHub, Docker, Streamlit, SQLAlchemy, Prisma, Vite

PROJECTS

1. **MedInsight AI — Medical Report Intelligence Platform**
   React, TypeScript, Tailwind CSS, FastAPI, Python, PostgreSQL, SQLAlchemy, Docker
   - Developed a full-stack platform that ingests uploaded medical lab reports and converts them into structured, queryable data, using a React/TypeScript frontend and a FastAPI backend.
   - Built an OCR and extraction pipeline that parses report text and routes it through separate analysis and knowledge services before returning results to the client.
   - Modeled the data layer in PostgreSQL with SQLAlchemy and handled schema changes through Alembic migrations.
   - Containerized the application with Docker to keep the frontend, backend, and database services isolated and reproducible.

2. **OcuVision AI — Retinal Disease Detection System**
   Python, PyTorch, EfficientNet-B0, FastAPI, Next.js, React, TypeScript, Grad-CAM
   - Trained an EfficientNet-B0 model in PyTorch on the ODIR retinal fundus dataset to classify eight categories of eye disease from a single image, framed as a multi-label classification task.
   - Added Grad-CAM visualizations so each prediction comes with a heatmap of the image regions the model weighted most heavily.
   - Built a FastAPI backend to handle image uploads and serve model predictions.
   - Built a Next.js/React dashboard for reviewing risk assessments and generated reports.

3. **SmartGridGuard — Explainable Electricity Theft Detection System**
   Python, LightGBM, Pandas, NumPy, Scikit-learn, SHAP, Streamlit
   - Built a model that scores utility customers by risk of electricity theft based on their historical consumption patterns.
   - Engineered 88 statistical, behavioral, temporal, and consumption-based features using Pandas and NumPy.
   - Trained a LightGBM classifier and used SHAP to surface which features were driving each customer's risk score.
   - Built a Streamlit dashboard for filtering, ranking, and prioritizing customers for investigation.

4. **IntelliFlow AI — Intelligent Traffic Management & Signal Optimization**
   Python, YOLOv8, OpenCV, FastAPI, React, TypeScript, PostgreSQL
   - Developed a system that processes uploaded traffic video to estimate vehicle volume at an intersection.

- Used YOLOv8 Nano with OpenCV to detect and classify cars, motorcycles, buses, and trucks frame by frame.
- Added deduplication and directional-classification logic so vehicles are counted once and attributed to the correct direction of travel.
- Wrote a signal-timing algorithm that allocates green-light duration by traffic volume, with a FastAPI backend and React dashboard for reviewing results.

5. **FNMart — Full-Stack Grocery Delivery Platform**
   Flutter, Dart, Node.js, Express.js, GetX, Razorpay
   - Built a grocery delivery app in Flutter covering product browsing, categories, cart, checkout, order history, and account management.
   - Built REST APIs with Node.js and Express.js to handle the data and business logic behind the app.
   - Used GetX for state management and organized the codebase around an MVC structure.
   - Integrated Razorpay for in-app payments and built reusable Flutter components across screens.

EDUCATION

Bachelor of Engineering (B.E.) — Computer Science Engineering (Artificial Intelligence & Machine Learning)
M.H. Saboo Siddik College of Engineering, University of Mumbai
2026 | CGPA: 7.55/10

RELEVANT COURSEWORK

Data Structures & Algorithms | Database Management Systems | Computer Networks | Operating Systems | Artificial Intelligence | Machine Learning | Deep Learning | Social Media Analysis | Image & Video Processing | Full Stack Development

LANGUAGES

English | Hindi | Marathi